

ALFAAJ63710

## Proteinase K, Ready-to-Use solution

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:**  
**Product Description:** 蛋白酶 K, 备用 溶液  
Proteinase K, Ready-to-Use solution

**Cat No. :** J63710

**Supplier** Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
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Lancashire, LA3 2XY,  
United Kingdom  
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**Emergency Telephone Number** For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
No information available

#### Emergency Overview

Causes mild skin irritation. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause respiratory irritation.

#### Classification of the substance or mixture

|                           |            |
|---------------------------|------------|
| Skin Corrosion/Irritation | Category 3 |
| Respiratory Sensitization | Category 1 |

#### Label Elements



**Signal Word**

**Danger**

#### Hazard Statements

H316 - Causes mild skin irritation

H334 - May cause allergy or asthma symptoms or breathing difficulties if inhaled

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H335 - May cause respiratory irritation

### Precautionary Statements

#### Prevention

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray

P284 - In case of inadequate ventilation wear respiratory protection

#### Response

P304 + P341 - IF INHALED: If breathing is difficult, remove victim to fresh air and keep at rest in a position comfortable for breathing

P342 + P311 - If experiencing respiratory symptoms: Call a POISON CENTER or doctor

#### Storage

P403 - Store in a well-ventilated place

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Physical and Chemical Hazards

None identified.

### Health Hazards

Causes mild skin irritation. May cause allergy or asthma symptoms or breathing difficulties if inhaled. May cause respiratory irritation.

### Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                                  | CAS No     | Weight % |
|------------------------------------------------------------|------------|----------|
| Glycerin                                                   | 56-81-5    | 50       |
| Water                                                      | 7732-18-5  | 47.82    |
| Proteinase, tritirachium album serine                      | 39450-01-6 | 2        |
| 1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, hydrochloride | 1185-53-1  | 0.16     |
| Calcium acetate                                            | 62-54-4    | 0.02     |

## SECTION 4. FIRST AID MEASURES

### General Advice

If symptoms persist, call a physician.

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

### Ingestion

Clean mouth with water and drink afterwards plenty of water.

### Most important symptoms and effects

May cause allergy or asthma symptoms or breathing difficulties if inhaled. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

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**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions**

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage**

Store in freezer.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters**

| Component | ACGIH TLV | OSHA PEL                                                                                                                           | NIOSH | The United Kingdom                         | European Union |
|-----------|-----------|------------------------------------------------------------------------------------------------------------------------------------|-------|--------------------------------------------|----------------|
| Glycerin  |           | (Vacated) TWA: 10 mg/m <sup>3</sup><br>(Vacated) TWA: 5 mg/m <sup>3</sup><br>TWA: 15 mg/m <sup>3</sup><br>TWA: 5 mg/m <sup>3</sup> |       | TWA: 10 mg/m <sup>3</sup> 8 hr (mist only) |                |

Legend

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OSHA - Occupational Safety and Health Administration

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

#### Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particulates filter conforming to EN 143

**Small scale/Laboratory use** Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.  
**Recommended half mask:-** Particle filtering: EN149:2001  
 When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** No information available.

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                 |                          |                                          |
|---------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>               | Colorless                |                                          |
| <b>Physical State</b>           | Liquid                   |                                          |
| <b>Odor</b>                     | No information available |                                          |
| <b>Odor Threshold</b>           | No data available        |                                          |
| <b>pH</b>                       | 8                        |                                          |
| <b>Melting Point/Range</b>      | No data available        |                                          |
| <b>Softening Point</b>          | No data available        |                                          |
| <b>Boiling Point/Range</b>      | No information available |                                          |
| <b>Flash Point</b>              | No information available | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | No data available        |                                          |
| <b>Flammability (solid,gas)</b> | Not applicable           | Liquid                                   |

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|                                                |                          |             |
|------------------------------------------------|--------------------------|-------------|
| <b>Explosion Limits</b>                        | No data available        |             |
| <b>Vapor Pressure</b>                          | No data available        |             |
| <b>Vapor Density</b>                           | No data available        | (Air = 1.0) |
| <b>Specific Gravity / Density</b>              | No data available        |             |
| <b>Bulk Density</b>                            | Not applicable           | Liquid      |
| <b>Water Solubility</b>                        | Miscible                 |             |
| <b>Solubility in other solvents</b>            | No information available |             |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |             |
| <b>Component</b>                               | <b>log Pow</b>           |             |
| Glycerin                                       | -1.75                    |             |
| 1,3-Propanediol,                               | -3.6                     |             |
| 2-amino-2-(hydroxymethyl)-, hydrochloride      |                          |             |
| <b>Autoignition Temperature</b>                | No data available        |             |
| <b>Decomposition Temperature</b>               | No data available        |             |
| <b>Viscosity</b>                               | No data available        |             |
| <b>Explosive Properties</b>                    | No information available |             |
| <b>Oxidizing Properties</b>                    | No information available |             |

### SECTION 10. STABILITY AND REACTIVITY

|                                 |                                 |
|---------------------------------|---------------------------------|
| <b>Stability</b>                | Stable under normal conditions. |
| <b>Hazardous Reactions</b>      | None under normal processing.   |
| <b>Hazardous Polymerization</b> | No information available.       |
| <b>Conditions to Avoid</b>      | None known.                     |
| <b>Materials to avoid</b>       | No information available.       |

**Hazardous Decomposition Products** Nitrogen oxides (NOx). Sulfur oxides. Hydrogen chloride. Calcium oxides.

### SECTION 11. TOXICOLOGICAL INFORMATION

#### Product Information

##### (a) acute toxicity; Toxicology data for the components

| Component                                 | LD50 Oral            | LD50 Dermal                   | LC50 Inhalation              |
|-------------------------------------------|----------------------|-------------------------------|------------------------------|
| Glycerin                                  | 12600 mg/kg ( Rat )  | > 10 g/kg ( Rabbit )          | > 2.75 mg/L/4h ( Rat )(mist) |
| Water                                     | -                    | -                             | -                            |
| 1,3-Propanediol,                          | OECD 425 (Rat)       | OECD 402 (Rat)                |                              |
| 2-amino-2-(hydroxymethyl)-, hydrochloride | LD50 > 5000 mg/kg bw | LD50 > 5000 mg/kg bw          |                              |
| Calcium acetate                           | 4280 mg/kg (Rat)     | LD50 > 20000 mg/kg ( Rabbit ) | LC50 > 5.6 mg/L ( Rat ) 4 h  |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

|                    |                   |
|--------------------|-------------------|
| <b>Respiratory</b> | Category 1        |
| <b>Skin</b>        | No data available |

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| Component                                                                           | Test method             | Test species | Study result    |
|-------------------------------------------------------------------------------------|-------------------------|--------------|-----------------|
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride<br>1185-53-1 ( 0.16 ) | OECD Test Guideline 406 | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

| Component                                                                           | Test method                                                | Test species          | Study result |
|-------------------------------------------------------------------------------------|------------------------------------------------------------|-----------------------|--------------|
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride<br>1185-53-1 ( 0.16 ) | OECD Test Guideline 471<br>Bacterial Reverse Mutation Test | Mammalian<br>in vitro | negative     |

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; No data available

**Symptoms / effects, both acute and delayed** Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

## SECTION 12. ECOLOGICAL INFORMATION

Ecotoxicity effects .

| Component                                                     | Freshwater Fish                                            | Water Flea                            | Freshwater Algae | Microtox                          |
|---------------------------------------------------------------|------------------------------------------------------------|---------------------------------------|------------------|-----------------------------------|
| Glycerin                                                      | LC50: 51 - 57 mL/L,<br>96h static<br>(Oncorhynchus mykiss) |                                       |                  |                                   |
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride |                                                            | Daphnia Magna<br>EC50 >100 mg/L (48h) |                  | OECD 209<br>EC50 > 1000 mg/L (3h) |

### Persistence and Degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential** Bioaccumulation is unlikely

| Component                                                     | log Pow | Bioconcentration factor (BCF) |
|---------------------------------------------------------------|---------|-------------------------------|
| Glycerin                                                      | -1.75   | No data available             |
| 1,3-Propanediol,<br>2-amino-2-(hydroxymethyl)-, hydrochloride | -3.6    | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

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**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

Not Regulated

**IMDG/IMO**

Not regulated

**IATA**

Not regulated

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component                                                  | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL     |
|------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|----------|
| Glycerin                                                   | -                                                   | -                                       | X    | X     | 200-289-5 | X    | X   | X     | X    | X    | X    | KE-29297 |
| Water                                                      | -                                                   | -                                       | X    | X     | 231-791-2 | X    | X   | X     | X    |      | X    | KE-35400 |
| Proteinase, tritirachium album serine                      | -                                                   | -                                       | X    | X     | 254-457-8 | -    | -   | -     | -    | X    | -    | -        |
| 1,3-Propanediol, 2-amino-2-(hydroxymethyl)-, hydrochloride | -                                                   | -                                       | X    | X     | 214-684-5 | X    | X   | X     | X    |      | X    | KE-34819 |
| Calcium acetate                                            | -                                                   | -                                       | X    | X     | 200-540-9 | X    | X   | X     | X    | X    | X    | KE-04463 |

**National Regulations****SECTION 16. OTHER INFORMATION**

Prepared By  
Revision Date

Health, Safety and Environmental Department  
09-May-2024

# SAFETY DATA SHEET

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### Revision Summary

New emergency telephone response service provider.

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**